Low- and high-level information analyses of transcriptome connecting endometrial-decidua-placental origin of preeclampsia subtypes: A preliminary study

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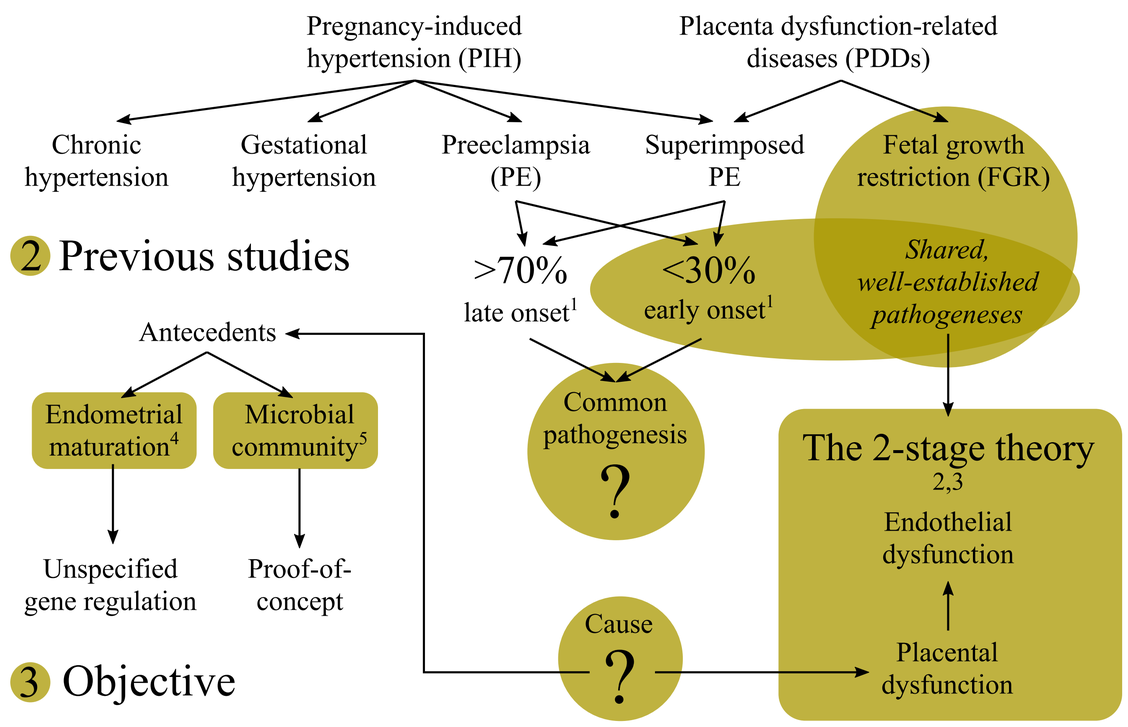
Background

1 Problem statement



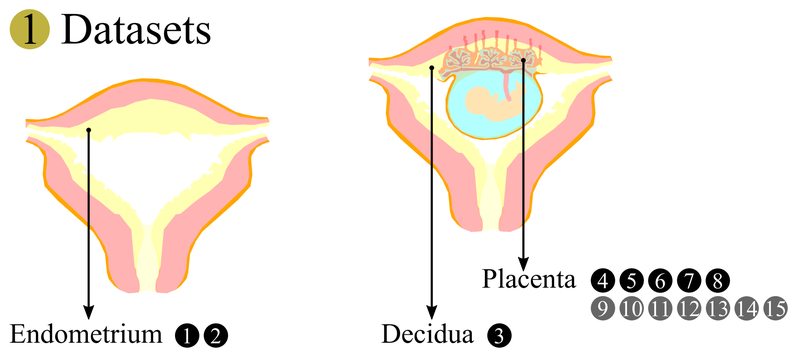
2 Previous studies

3 Objective

To identify transcriptome representing endometrial maturation and histologic chorioamnionitis, enriched by differentially expressed genes of the preeclampsia subtypes, using microarray meta-analysis workflow at low- and high-level information.

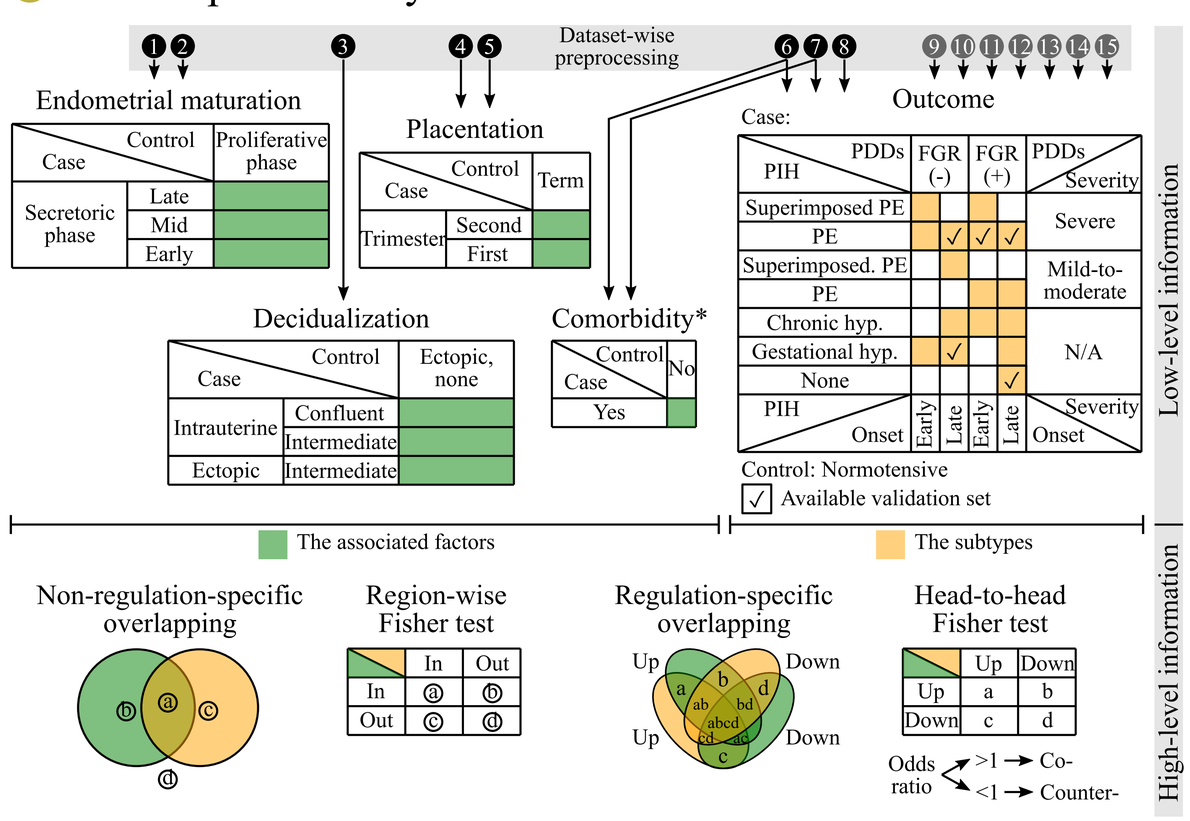
Methods

1 Datasets



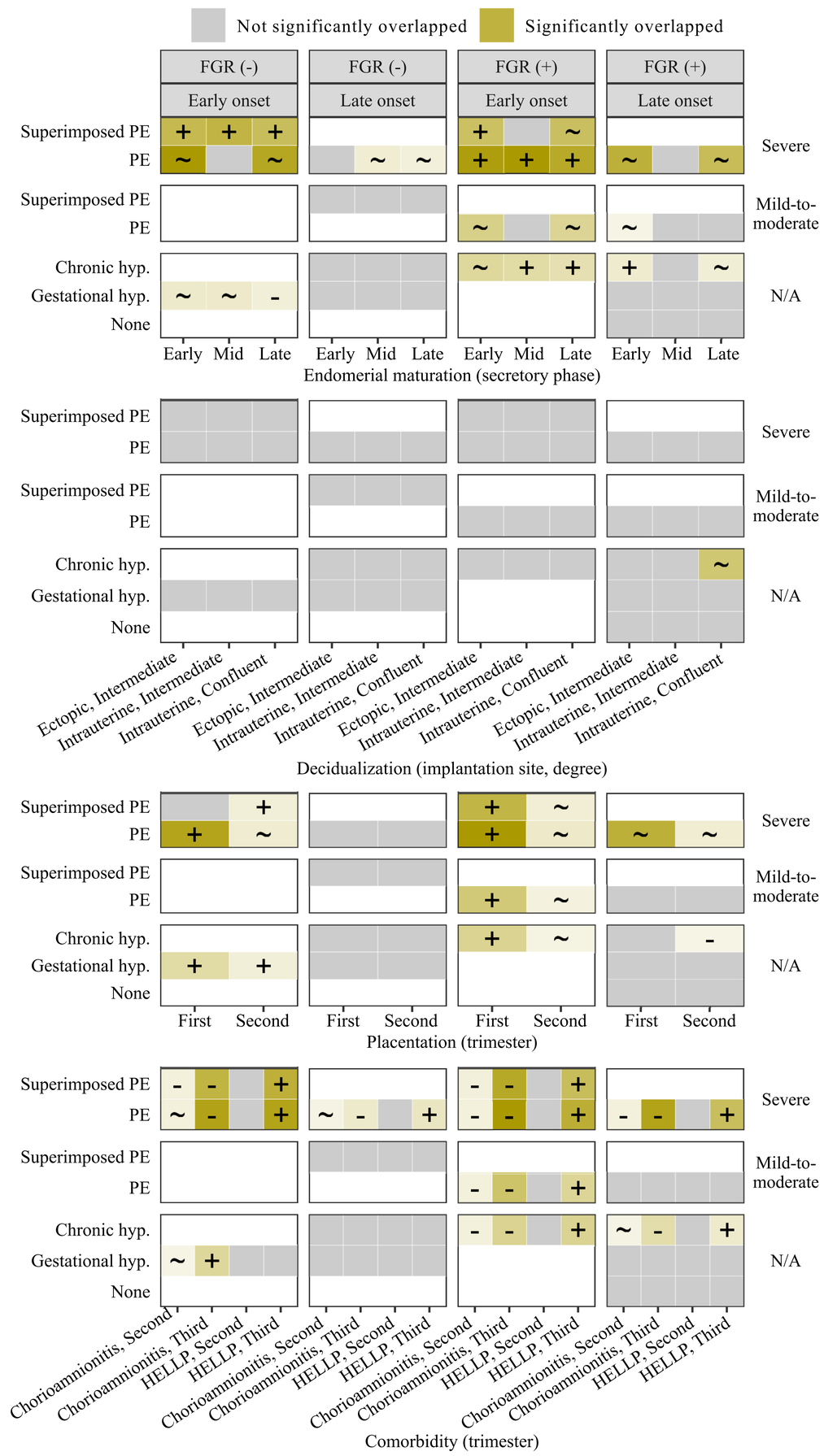
Discovery sets	Validation sets
❶ GSE4888	❾ GSE30186
❷ GSE6364	❿ GSE10588
❸ E-MTAB-680	⓫ GSE24129
❹ GSE12767	⓬ GSE25906
❺ GSE9984	⓭ GSE4707
❻ GSE75010	⓮ GSE44711
❼ GSE98224	⓯ GSE128381
❽ GSE100415	

2 Transcriptome analysis



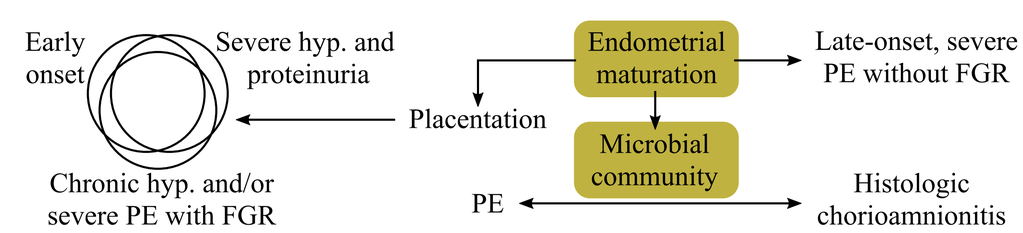
*, conducted for chorioamnionitis/HELLP in either second or third trimester; FGR, fetal growth restriction; HELLP, hemolysis, elevated liver enzymes, and low platelets; hyp.; hypertension; N/A, not applicable; PDDs, placenta dysfunction-related diseases; PE, preeclampsia; PIH, pregnancy-induced hypertension.

Results



Significances of undifferentiated, co-, and counter-expression are respectively indicated by ~, +, and -. The color gradation represents the number of overlapped DEGs for undifferentiated expression and the number of either co- or counter-expressed DEGs. All non-grey tiles are significant for the non-regulation-specific overlapping. DEGs, differentially-expressed genes; FGR, fetal growth restriction; hyp.; hypertension; PE, preeclampsia; N/A, not applicable.

Conclusions



In addition, our preliminary results showed the feasibility of developing and validating pathogenesis models of PE subtypes which will be the focus of the extension work of this preliminary study. Eventually, this study will help to decide if future studies need prospective, pre-pregnancy and chorioamnionitis data for preeclampsia.

SCAN ME

https://qrco.de/beRSk0

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